CHAPTER 3

Nuclei, Isotopes and Isotope Separation

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From a practical point of nuclear chemistry we accept the commonly used model that the nucleus is composed of only protons and neutrons. For the purposes of the scope of this book this definition is enough even if, as seen in Chapter 2, the world is far more complex than that. Although this is a simplification most observable patters of nuclear chemistry can be explained and the nomenclature can be kept simple. For example, the oxygen atom of mass number 16 has a nucleus which consists of 8 protons and 8 neutrons; since neutrons have no charge but are very similar to protons in mass, the net nuclear charge is +8. There are also 8 extranuclear electrons in the neutral atom of oxygen.

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3.1. SPECIES OF ATOMIC NUCLEI

Both the protons and the neutrons in the nucleus are commonly denominated *nucleons*. The *mass number A* is the total number of nucleons. Thus

$$A = N + Z \tag{3.1}$$

where Z is the number of protons, i.e. the *atomic number* and N is the number of neutrons. The elemental identity and the chemical properties are basically determined by the atomic number even if most chemical interactions take place via the interactions of the surrounding electron cloud.

As we have seen in Chapter 1, an element may be composed of atoms that, while having the same number of protons in the nuclei, have different mass numbers and, therefore, different numbers of neutrons. Neon, for example, has an atomic number of 10, which means that the number of protons in the nuclei of all neon atoms is 10; however, 90% of the neon atoms in nature have 10 neutrons present in their nuclei while 10% of the atoms have 12 neutrons. Such atoms of constant Z but different A are called *isotopes*. The heavy hydrogen isotopes ^{2}H and ^{3}H are used so often in nuclear science that they have been given special names and symbols, deuterium (D) and tritium (T), respectively.

The word isotope is often misused to designate any particular nuclear species, such as ^{16}O, ^{14}C, ^{12}C. It is correct to call ^{12}C and ^{14}C isotopes of carbon since they are nuclear species of the same element. However, ^{16}O and ^{12}C are not isotopic since they belong to different elemental species. The more general word *nuclide* is used to designate any specific nuclear species; e.g. ^{16}O, ^{14}C, and ^{12}C are nuclides. The term *radionuclide* should be used to designate any radioactive nuclide, although, somewhat illogically, *radioisotope* is a common term used for the same purpose.

In addition to being classified into isotopic groups, nuclides may also be divided into groupings with common mass numbers and common neutron numbers. *Isotopes* are nuclides with a common number of protons (Z), whereas *isobar* is the term used to designate nuclides with a common number of nucleons (A), i.e. the same mass number. Nuclei with the same number of neutrons (N) but different atomic numbers are termed *isotones* ^{40}K and ^{40}Ar are examples of isobars, while ^{3}H and ^{4}He are examples of isotones.

In some cases a nucleus may exist for some time in one or more excited states and it is differentiated on this basis. Such nuclei that necessarily have the same atomic number and mass number are called *isomers*. ^{60m}Co and ^{60g}Co are isomers; the ^{60m}Co nuclide exists in a high energy (*excited*) *state* and decays spontaneously by emission of a γ-ray with a half-life of 10.5 min to the lowest energy, *ground state*, designated by ^{60g}Co.

$$^{60m}\mathrm{Co} \xrightarrow[10.5\ \mathrm{min}]{\gamma} {}^{60g}\mathrm{Co} \tag{3.2}$$

The symbol m stands for *metastable*, while g (or no symbol) refers to the ground state. In order for a nuclide to be called *metastable* one usually demand that the excited state has a half-live of at least 10^{-9} s. This is about 1000 times longer than for a normal excited